

MINGZHE DU

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Innovation 4.0, 3 Research Link, Singapore, 117602

RESEARCH INTERESTS

Self-evolving Harness / Long-horizon Code Generation / Preference Alignment

EDUCATION

Nanyang Technological University

PhD Candidate in Computer Science, College of Computing and Data Science

Singapore

Aug. 2022 – Present

- Advisor: *Prof. Anh Tuan Luu*

The University of Melbourne

Master of Information Technology, Faculty of Engineering

Melbourne, Australia

Oct. 2017 – Dec. 2019

- Advisor: *Prof. Richard O. Sinnott*
- Specialization: Distributed Computing (Honorable Dissertation)

Central South University

Bachelor of Information Security, School of Computer Science and Engineering

Changsha, China

Sep. 2013 – Jun. 2017

- Excellent Scholarships (*GPA: 3.7/4.0*)

PROFESSIONAL EXPERIENCE

TikTok, *AI Innovation Center, Singapore*

May 2026 – Present

- Research Scientist (Full Time / Soaring Star)
- Developing AI systems for large-scale, repository-level code efficiency optimization, and bridging research and practice by deploying them to optimize production codebases.

National University of Singapore, *Institute of Data Science*

Aug. 2021 – May 2026

- Research Associate (Advisor: *Prof. See-Kiong Ng*)
- Affiliated with Cisco-NUS Accelerated Digital Economy Corporate Laboratory.

ByteDance, *TikTok Search, Beijing*

Dec. 2019 – Aug. 2021

- Software Architect (Full Time)
- Architected a scalable, general-purpose search engine (*Search Eternal*) that now powers core search services for major ByteDance platforms, including *Douyin*, *TikTok* and *Toutiao*.

ACADEMIC SERVICES

- **Conference Reviewer:** *NeurIPS (24,25,26), ICLR (24,25,26,27), ICML (25,26), AISTATS (24,26), AAAI (26,27), CIKM (25,26), AISI'26, ECAI'25, NAACL'24, EMNLP'24, ECAI'24, CLEF'23.*
- **Journal Reviewer:** *ACM Computing Survey, Frontiers in Artificial Intelligence.*
- **Volunteer:** *NeurIPS'25, ACL'25, WWW'24, EMNLP'24, AAAI'23.*
- **STEM Mentor:** *The University of Melbourne (2019-Present).*

SELECTED PUBLICATIONS

[**NeurIPS'25**] [Afterburner: Reinforcement Learning Facilitates Self-Improving Code Efficiency Optimization.](#)

Mingzhe Du, Luu Anh Tuan, Yue Liu, Yuhao Qing, Dong Huang, Xinyi He, Qian Liu, Zejun Ma, See-kiong Ng

[**ACL'25**] [CodeArena: A Collective Evaluation Platform for LLM Code Generation.](#)

Mingzhe Du, Luu Anh Tuan, Bin Ji, Xiaobao Wu, Dong Huang, Terry Yue Zhuo, Qian Liu, See-Kiong Ng

[**NeurIPS'24**] [Mercury: A Code Efficiency Benchmark for Code Large Language Models.](#)

Mingzhe Du, Luu Anh Tuan, Bin Ji, Qian Liu, See-Kiong Ng

- [**Preprint**] [Pareto-Gated Credit Assignment for Concise Unit-Test Generation with Reinforcement Learning](#). Dong Huang, Mark Harman, Jie M. Zhang, Zhijiang Guo, **Mingzhe Du**[†], See-Kiong Ng.
- [**Preprint**] [Measuring the Checker: Mutation Analysis for GPU-Kernel Benchmark Oracles](#). Dong Huang, Tianyi Wu, Xuan Zhang, Renyang Liu, Luu Anh Tuan, See-Kiong Ng, **Mingzhe Du**[†].
- [**Preprint**] [Mastermind: Strategy-Level Learning for Repository-Scale Vulnerability Reproduction](#). **Mingzhe Du**, Luu Anh Tuan, Dong Huang, Renyang Liu, Tianyi Wu, Zhijiang Guo, See-kiong Ng.
- [**Preprint**] [Secure Code Generation via Online Reinforcement Learning with Vulnerability Reward Model](#). Tianyi Wu, **Mingzhe Du**, Yue Liu, Chengran Yang, Terry Yue Zhuo, Jiaheng Zhang, See-Kiong Ng.
- [**Preprint**] [Scaling Code LLM Training and Test-Time Inference via Execution-Free Reward Models](#). Xiao Zhu, Xinyu Zhou, Boyu Zhu, Hanxu Hu, **Mingzhe Du**, Haotian Zhang, Huiming Wang, Zhijiang Guo.
- [**CIKM'26**] [Paper Espresso: From Paper Overload to Research Insight. \(Oral and Best Paper Nomination\)](#) **Mingzhe Du**, Luu Anh Tuan, Dong Huang, See-kiong Ng.
- [**ICML'26**] [SWE-Perf: Can Language Models Optimize Code Performance on Real-World Repositories?](#) Xinyi He, Qian Liu, **Mingzhe Du**, Lin Yan, Zhijie Fan, Yiming Huang, Zejian Yuan, Zejun Ma.
- [**ICML'26 DL4C**] [Nexus: Execution-Grounded Multi-Agent Test Oracle Synthesis](#) Dong Huang, **Mingzhe Du**[†], Jie M. Zhang, Zheng Lin, Meng Luo, Qianru Zhang, See-kiong Ng.
- [**ACL'26**] [Semantics-Aligned, Curriculum-Driven, and Reasoning-Enhanced Vulnerability Repair Framework \(Oral and Best Paper Nomination\)](#) Chengran Yang, Ting Zhang, Jinfeng Jiang, Xin Zhou, Haoye Tian, **Mingzhe Du**, Jieke Shi, Junkai Chen, Yikun Li, Eng Lieh Ouh, Lwin Khin Shar, David Lo.
- [**TOSEM'26**] [Benchmarking LLMs for Unit Test Generation from Real-World Functions](#). Dong Huang, Jie M. Zhang, Mark Harman, Qianru Zhang, **Mingzhe Du**[†], See-Kiong Ng.
- [**EACL'26**] [Pro-QuEST: Prompt-chaining Quiz Engine for testing Specialized Technical Product Knowledge](#) Sujatha Das Gollapalli, Mouad Hakam, **Mingzhe Du**, See-Kiong Ng, Mohammed Hamzeh.
- [**AMIYA'26**] [Improving Arabic Dialectness in LLMs with Reinforcement Learning](#) Sujatha Das Gollapalli, Mouad Hakam, **Mingzhe Du**, See-Kiong Ng, Mohammed Hamzeh.
- [**ICLR'26**] [Beyond Prompt-Induced Lies: Investigating LLM Deception on Benign Prompts. \(Oral\)](#) Zhaomin Wu, **Mingzhe Du**, See-Kiong Ng, Bingsheng He.
- [**EMNLP'25**] [On Assigning Product and Software Codes to Service Requests with Large Language Models](#). Sujatha Das Gollapalli, Mouad Hakam, **Mingzhe Du**, See-Kiong Ng, Mohammed Hamzeh.
- [**NeurIPS'25**] [EffiBench-X: A Multi-Language Benchmark for Measuring Efficiency of LLM-Generated Code](#). Yuhao Qing*, Boyu Zhu*, **Mingzhe Du***, Zhijiang Guo, Terry Yue Zhuo, Qianru Zhang, Jie M. Zhang, Heming Cui, Siu-Ming Yiu, Dong Huang, See-Kiong Ng, Luu Anh Tuan.
- [**NeurIPS'25**] [Guardreasoner-VL: Safeguarding VLMs via Reinforced Reasoning](#). Yue Liu, Shengfang Zhai, **Mingzhe Du**, Yulin Chen, Tri Cao, Hongcheng Gao, Cheng Wang, Xinfeng Li, Kun Wang, Junfeng Fang, Jiaheng Zhang, Bryan Hooi.
- [**ICSE'25**] [Measuring the Influence of Incorrect Code on Test Generation](#). Dong Huang, Jie M. Zhang, Mark Harman, **Mingzhe Du**[†], and Heming Cui.
- [**ACL'25**] [AntiLeakBench: Preventing Data Contamination by Automatically Constructing Benchmarks with Updated Real-World Knowledge. \(Oral and SAC Highlight Award\)](#) Xiaobao Wu, Liangming Pan, Yuxi Xie, Ruiwen Zhou, Shuai Zhao, Yubo Ma, **Mingzhe Du**, Rui Mao, Anh Tuan Luu, and William Yang Wang.
- [**ICML'25**] [Current Model Licensing Practices are Dragging Us into a Quagmire of Legal Noncompliance. \(Oral\)](#) Moming Duan*, **Mingzhe Du***, Rui Zhao, Mengying Wang, Yinghui Wu, Nigel Shadbolt, and Bingsheng He.

[AAAI'25] [Towards Verifiable Text Generation with Generative Agent. \(Oral\)](#)

Ji, Bin, Huijun Liu, **Mingzhe Du**, Shasha Li, Xiaodong Liu, Jun Ma, Jie Yu, and See-Kiong Ng.

[SAC'25] [Curriculum Demonstration Selection for In-Context Learning.](#)

Duc Anh Vu, Cong-Duy Nguyen, Xiaobao Wu, Nhat Hoang, **Mingzhe Du**, Thong Nguyen, and Anh Tuan Luu.

[JMIR'25] [Unraveling Online Mental Health Through the Lens of Early Maladaptive Schemas: AI-Enabled Content Analysis of Online Mental Health Communities.](#)

Beng Heng Ang, Sujatha Das Gollapalli, **Mingzhe Du**, and See-Kiong Ng.

[AAAI'24] [From Static to Dynamic: Knowledge Metabolism for Large Language Models.](#)

Mingzhe Du, Anh Tuan Luu, Bin Ji, and See-Kiong Ng.

[AAAI'24] [Chain-of-Thought Improves Text Generation with Citations in Large Language Models.](#)

Bin Ji, Huijun Liu, **Mingzhe Du**, and See-Kiong Ng.

[ECAI'24] [Counseling Responses for Mental Health Forum Questions with Maladaptive Schema Prediction.](#)

Sujatha Das Gollapalli, Beng Heng Ang, **Mingzhe Du**, and See-Kiong Ng.

[TLDK'23] [Constituency-Informed and Constituency-Constrained Extractive Question Answering with Heterogeneous Graph Transformer.](#)

Mingzhe Du, Mouad Hakam, See-Kiong Ng, and Stéphane Bressan.

[WWW'23] [Identifying Checkworthy Cure Claims on Twitter.](#)

Sujatha Das Gollapalli, **Mingzhe Du**, and See-Kiong Ng.

[AAAI'23] [Generating Reflective Questions for Engaging Gallery Visitors in ArtMuse.](#)

Sujatha Das Gollapalli, **Mingzhe Du**, and See-Kiong Ng.

[CLEF'22] [NUS-IDS at CheckThat! 2022: Identifying Check-worthiness of Tweets using CheckthaT5.](#)

Mingzhe Du, Sujatha Das Gollapalli, and See-Kiong Ng.

† *Corresponding Author*, * *Equal Contribution*.

AWARDS & HONORS

- **NVIDIA Academic Grant:** *Predicting and Optimizing LLVM Compiler Pass Order in a Unified Model.*
- **Thinking Machines Lab Research Grant:** *Scaling Reinforcement Learning for Efficient Code Generation.*
- **NVIDIA Academic Grant:** *Scaling Laws of Reinforcement Learning for Code Efficiency Optimization.*
- **Staff Research Award** at *Institute of Data Science, National University of Singapore, 2025.*
- **Google Academic Research Award:** *SafeCodeX: Security-Aware Code Generation with LLMs, 2025.*
- **First Runner-Up Award** at the **Singapore Healthcare AI Datathon:** *Identifying ICU Patient with Central Lines at Risk of CLABSI, 2024.*
- **First Prize** at **Catalyst:** *A Path Recommendation System based on Data Aggregation and Analysis, 2019.*
- **First Prize** at **Codebrew:** *After Dark: Finding Safe and Fast Routes for Pedestrians at Night, 2019.*
- **Technology Pioneer Award** at the *Cybersecurity Talent Development and Entrepreneurship Forum, 2015.*
- **First Prize** at the *Mathematical Contest in Modeling: Study on the Wreckage Position of Malaysia Airlines Flight MH370, 2014.*

MISCELLANEOUS

- **LinkedIn:** <https://www.linkedin.com/in/dumingzhe> **WeChat:** *spirit_song*
- **GitHub:** <https://www.github.com/elfsong> **Language:** *Mandarin (native) | English (professional)*